Contrastive Retrieval Augmented In-Context Learning for Medical Classification Tasks with Imbalanced Data

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***Abstract*—Pre-trained large language models (LLMs) can be used as classifiers by using in-context learning where we provide a few shot examples along with task description in the prompt. We study the use of LLMs for classification tasks in the medical domain, where class imbalance is common - for example, predicting whether a patient develops a particular disease or whether (s)he experiences a treatment-related side effect. Although providing a few-shot examples of similar patients generally improves classification accuracy, unbalanced datasets often cause retrieved neighbors to be dominated by the majority (negative) class that can lead to a bias towards predicting the negative class. To address this, we propose C-RAIL (Contrastive Retrieval-Augmented In-Context Learning), a method that retrieves neighboring instances from both classes and incorporates them as balanced few-shot examples. This mitigates the class imbalance problem by ensuring contrastive representation of positive and negative outcomes in the prompt. Experiments across two medical domain datasets show that C-RAIL consistently improves predictive performance compared to zero shot as well as standard retrieval-based in-context learning approach. This shows that LLMs, combined with contrastive retrieval in context learning can be effectively used for classification tasks in medical domain.**

***Index Terms*—In context learning, Classification, Imbalanced data**

I. Introduction

Large Language Model (LLMs) have shown tremendous improvements in tasks that require generation of language such as machine translation [1], and question answering [2]. They have also demonstrated strong performance across a wide variety of tasks when used as a classifier [3]. Recent work has shown that LLMs, without additional fine-tuning, can achieve competitive results in classification tasks through *in-context learning* (ICL), where predictions are guided by task instructions and a few illustrative examples provided directly in the prompt [3], [4].

The choice of in-context examples significantly effects the overall classification performance [13] and therefore researchers have explored retrieval paradigm to determine which few shot examples should be used for a given input. In this paradigm, for a given input instance, semantically similar examples are retrieved from a dataset and retrieved examples are augmented into the prompt as few-shot demonstrations. In this paper, we refer to this approach as Retrieval Augmented in-context learning or RAIL. Empirical studies have shown that such retrieval can significantly boost classification accuracy and robustness across a variety of tasks [5]–[7].

The ability of in-context learning makes LLMs an excellent choice for classification tasks where labeled data is limited or expensive to acquire such as in the medical domain. However, applying these methods in the medical domain presents unique challenges. In medical prediction tasks, such as forecasting disease outcomes or identifying treatment-related side effects, datasets are frequently affected by severe class imbalance. In such cases, naive retrieval strategies often return neighbors that overwhelmingly belong to the majority class. This can lead to an unnecessary bias towards predicting the majority class. This issue is particularly problematic, where the minority class (e.g., patients who develop a side effect) is often the most critical to detect [8]–[10].

In this paper, we propose Contrastive Retrieval-Augmented In-Context Learning (C-RAIL), a novel approach designed to address the effects of class imbalance in retrieved neighbors for a given test input. Rather than retrieving only the most similar neighbors, C-RAIL retrieves examples from both the classes and incorporates them together into the context in a balanced and contrastive manner. This ensures that LLMs are simultaneously exposed to similar positive outcomes as well as similar negative outcomes during prediction, strengthening their ability to distinguish between classes. Our experimental results on two datasets demonstrate that C-RAIL consistently improves classification accuracy in imbalanced medical prediction tasks compared to zero shot as well as the RAIL approach.

The rest of the paper is organized as follows. We present related work in Section II and then provide details of our approach in Section III. We present experimental evaluations in Section IV and finally provide conclusion and future work in Section V

II. Related Work

Large language models (LLMs) have shown strong few-shot and in-context learning (ICL) abilities, and have demonstrated competitive performance without fine tuning requirements [3]. Min et al. show that performance is highly sensitive to the provided in-context examples [4]. The work done by [7] show that selecting examples that are semantically similar to the query often yields consistent gains over random selection. Several works have proposed methods that retrieve similar examples to each query, showing that retrieval-augmented ICL improves downstream performance across tasks, including classification [6], [14]. These methods use embedding based similarity metrics to retrieve similar examples. Embedding based methods excel in finding semantically similar textual data but they often struggle with numerical data that appear as part of clinical data. To overcome this, in a recent work Ronando et.al. [19] proposed a hybrid Euclidean distance with Large Language Models (HED-LM) framework, that integrates numerical similarity filtering with LLM-based contextual evaluation to optimize example selection in sensor-based data for fewshot prompting.

In-context learning for classification has also been studied in the medical domain. Recent studies explore different techniques to optimize in-context learning in biomedical natural language processing by improving example selection [16]. Zhan et al. show that retrieving similar examples for disease classification with multimodal LLM results in consistent accuracy and F1 improvements [15].

III. Methodology

A. Problem Definition

We focus on a binary classification problem in the medical domain. Let $D = \{(x_i, y_i)\}_{i=1}^{N}$ denote a dataset of N patients, where x_i represents the set of clinical features for a patient i, and $y_i \in \{0, 1\}$ denotes a label that captures the outcome of interest (e.g., whether the patient develops a disease or treatment-related side effect). In typical medical related problems, this dataset is imbalanced, such that $\Pr(y = 1)$ is much lesser than $\Pr(y = 0)$. This makes the prediction of the minority class more challenging.

Given a new patient instance x_q, the goal is to predict $\hat{y}_q$ using a pre-trained Large Language Model (LLM) in an in-context learning (ICL) setup. In this study, we aim to use the pre trained LLMs out-of-the-box with zero shot and few shot settings without fine tuning them for the task.

B. In-Context Learning for Classification

LLMs are naturally suited for generative tasks such as machine translation and question answering. They have also been shown to learn new tasks very effectively by using a few shot examples as part of their prompt [3]. We can easily adapt LLMs for a classification task by constructing a prompt containing a task description and a small set of labeled examples. This approach of providing a few examples as part of prompt along with task instructions is called In Context Learning or ICL. As an example, a few-shot ICL prompt can be expressed as:

$$P = \{I\} \cup \{(x_1, y_1), (x_2, y_2), \ldots, (x_k, y_k)\} \cup \{x_q\} \quad (1)$$

where I denotes the task instruction. Each pair (x_i, y_i) is an example with x_i being a set of all the input features and y_i representing the corresponding class label. The symbol x_q denotes the query instance. The LLM processes the prompt P and generates $\hat{y}_q$. Figure 2 shows an example prompt with 2-shot examples.

Prior works [3], [4] have shown that ICL can provide competitive classification performance without fine-tuning the model. The few shot examples that are used for a query instance x_q could be randomly chosen. However, it has been shown that the overall performance of the classification is strongly dependent on the choice of a few shot examples that are used [13].

C. Retrieval-Augmented In-Context Learning (RAIL)

To improve the few shot examples that are used with a prompt retrieval-augmented methods [6], [7] have been proposed. These methods dynamically retrieve examples from a set of training data D that are *similar* to the given query instance x_q. Let $N_k(x_q)$ denote the set of k similar retrieved examples. The retrieved neighbors $N_k(x_q)$ are then inserted into the prompt as few-shot examples:

$$P_{\text{RAIL}} = \{I\} \cup \{(x_i, y_i) \,|\, x_i \in N_k(x_q)\} \cup \{x_q\}. \quad (2)$$

RAIL provides the LLM with a context that is similar and therefore more relevant for the given test input. The approach has been shown to improve classification accuracy [12]. However, in imbalanced datasets the retrieved set of examples are often dominated by the examples from the majority class which is often the negative class. This can lead to bias towards predicting the negative outcome.

D. Contrastive Retrieval-Augmented In-Context Learning (C-RAIL)

To mitigate the problem of bias towards the negative class in the retrieved similar examples, we propose Contrastive Retrieval-Augmented In-Context Learning (C-RAIL). In this approach we use contrastive retrieval strategy that retrieves similar examples for each class separately and then use them jointly as part of the few shot examples in the prompt. More formally, to retrieve k similar examples for a test instance x_q, we retrieve $\frac{k}{2}$ nearest neighbors from the positive class and similarly $\frac{k}{2}$ nearest neighbors for the negative class. Let P_q and N_q denote the set of retrieved examples from the respective classes. We then construct the prompt as follows:

$$P_{\text{C-RAIL}} = \{I\} \cup \{(x_i^+, y_i = 1)\}_{i=1}^{\frac{k}{2}} \cup \{(x_j^-, y_j = 0)\}_{j=1}^{\frac{k}{2}} \cup \{x_q\}. \quad (3)$$

Here x_i^+ denotes an example from the set P_q and x_i^- denotes an example from the set N_q.

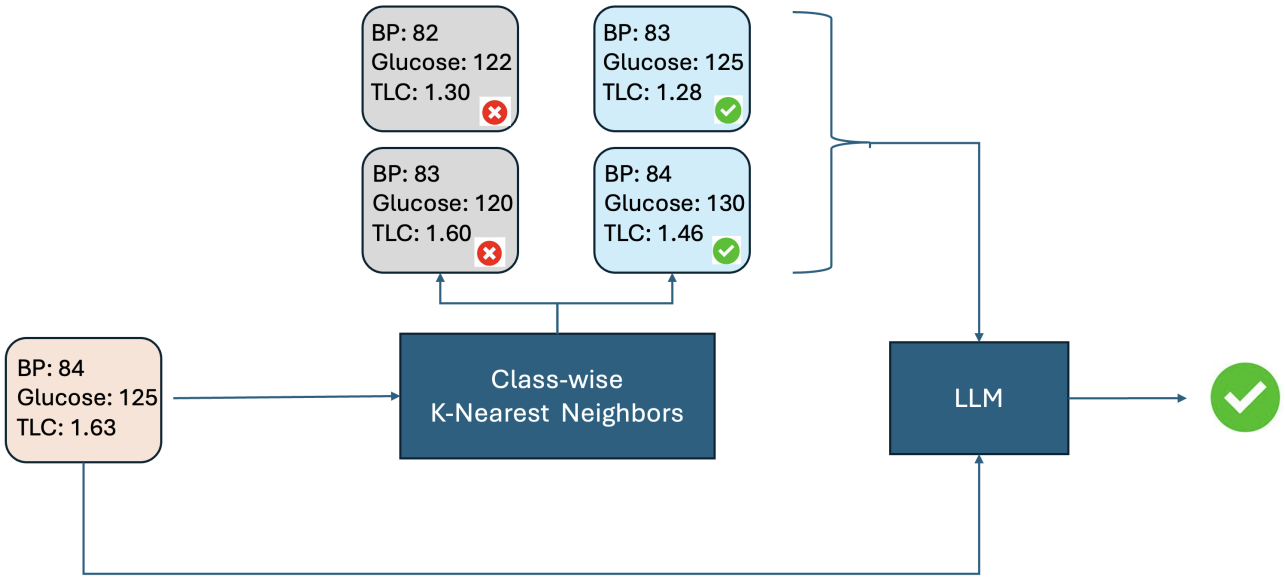


Fig. 1. Illustration of the proposed C-RAIL (Contrastive Retrieval-Augmented In-Context Learning) framework. A new patient's data is combined with retrieved balanced examples from both classes and fed into an LLM for prediction.

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You are a careful clinical classifier. Given a single
patient's anonymized features, predict whether the
patient has diabetes or not. Only use the provided
features—do not invent data.

Example 1.
Pregnancies: 1, Glucose: 148, BloodPressure: 74,
SkinThickness: 31, Insulin: 168, BMI: 33.6, DPF: 2.228,
Age: 50
Answer: 1

Example 2.
Pregnancies: 0,  Glucose: 137, BloodPressure: 78,
SkinThickness: 32, Insulin: 94, BMI: 28.1, DPF: 0.167,
Age: 28
Answer: 0

Now evaluate this patient
Pregnancies: 2,  Glucose: 159, BloodPressure: 72,
SkinThickness: 29, Insulin: 88, BMI: 27.4, DPF: 0.294,
Age: 40
```

Fig. 2. An example prompt for ICL with 2-shot examples.

Figure 1 provides an example scenario for the proposed approach. For a given patient represented by a set of key value pairs, we retrieve k examples for both the positive and negative class and then use them to constuct the prompt that is used by the LLM.

Algorithm 1: PreprocessData(X)

Input: Feature matrix X
Output: Processed feature matrix $\tilde{X}$

1) Compute the median, mean, and standard deviation for each column of X.
2) Replace missing values in each column with the corresponding column median.
3) Standardize each column using $z = (x - \mu)/\sigma$, where μ is the column mean and σ the column standard deviation.
4) Return $\tilde{X}$.

Fig. 3. Preprocessing with median imputation and z-score normalization.

We pre-process the data before determining the nearest neighbors. To deal with missing values in the data we use mean imputation method. In this method we compute mean of all the observed values for a feature column and then use that mean value in place of the missing values in the column. Since each column has widely different scales we standardize each column by first subtracting each value by the column mean and dividing it by the column standard deviation. We provide the details on pre processing of the data in Figure 3.

We use Euclidean distance for our similarity measure and compute distance with all the data points in the training dataset. Figure 4 provides details of this process. We use scikit-learn implementation for data pre-processing as well as for finding similar examples [18]. To find k similar example for

Algorithm 2: KNN_Retrieve(X_{train}, x_{query}, k)

Input: Training data X_{train}, query x_{query}, number of neighbors k

Output: Indices and distances of k nearest neighbors

1) $\tilde{X}_{train} \leftarrow$ PreprocessData(X_{train}).
2) $\tilde{x}_{query} \leftarrow$ PreprocessData(x_{query}).
3) For each row $\tilde{x}_i$ in $\tilde{X}_{train}$, compute the Euclidean distance $d_i = \|\tilde{x}_{query} - \tilde{x}_i\|_2$.
4) Sort all pairs (i, d_i) in ascending order of d_i.
5) Remove the query index from the list (if present).
6) Take the first k pairs and return (indices, distances).

Fig. 4. Retrieving k nearest neighbors after preprocessing (self is always removed).

the C-RAIL method, we call the function $KNN_Retriever$ given in Figure 4 twice. First we find closest $\frac{k}{2}$ positive examples by passing only the positive data points in training data set and then subsequently call this method again to find $\frac{k}{2}$ most similar negative examples by passing only the negative data points in the training dataset.

By providing the examples from both the classes, C-RAIL reduces the bias introduced by imbalance in the retrieval pool. The contrastive construction ensures that the LLM learns discriminative cues between positive and negative outcomes, improving its robustness in medical prediction tasks. Through experiments across multiple datasets, we show that C-RAIL consistently improves minority-class recall while maintaining balanced overall accuracy, outperforming both standard ICL and retrieval-augmented ICL.

IV. Experiments

A. Datasets

We evaluate our approach on two datasets:

1) Immunotherapy Dataset: The dataset consists of records from 118 patients who underwent immunotherapy for leukemia. Each record includes biochemical and clinical characteristics such as hemoglobin levels, platelet count, ferritin levels, and other laboratory measurements. The outcome variable indicates whether the patient developed severe immune-related side effects following therapy. Among the 118 patients, 18 experienced such side effects, representing approximately 15.3% of the cohort. This reflects a highly imbalanced class distribution.

2) Pima Indians Diabetes Dataset: The second dataset is the publicly available *Pima Indians Diabetes* dataset from the UCI repository [11]. It contains 768 instances with 8 clinical features such as number of pregnancies, body mass index, age, and plasma glucose concentration. The task is to predict the onset of diabetes within five years. The dataset is moderately imbalanced, with 268 positive cases (34.9%). Table I summarizes the statistics of both datasets used in our experiments.

TABLE I
Dataset statistics.

Dataset	# Instances	# Features	Positive Class (%)
Immunotherapy	118	28	15.3
Pima Diabetes	768	8	34.9

B. Evaluation Metrics

To assess the performance of different methods, we use 3 standard evaluation metrics namely precision, recall and F1-score. These metrics are defined as follows:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{4}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{5}$$

$$F_1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{6}$$

For the positive class, TP is the number of correctly predicted positive cases, FP is the number of cases that have been predicted to be positive but they belong to the negative class and FN represents the number of cases that have been predicted to be negative but they belong to the positive class. For both our tasks, the positive class (side effect following immunotherapy, diabetes onset in the Pima dataset) is of greater clinical interest. Detecting these outcomes with high recall is essential, as missing positive cases can have severe consequences. At the same time, high precision is necessary to avoid unnecessary concern and interventions. To capture this trade-off, we report the precision, recall and F1-score on the positive class as our primary evaluation metric.

C. Experimental Setup

For retrieval, we employ the k-nearest neighbors algorithm with Euclidean distance, after normalizing all feature values as described in Section III. For each query instance, we use $k = 6$ demonstrations in the prompt. In the case of C-RAIL, this set is balanced by including $k_p = 3$ positive-class and $k_n = 3$ negative-class examples.

We use the `Llama-3.1-8B` [17] model as the backbone LLM for all experiments. We use greedy decoding with 0 temperature to generate predictions. We compare three settings:

1) **Zero-shot:** In this setting we use task description without using any few shot examples.
2) **RAIL:** In this setting, given a test query x_q we retrieve k similar examples using a nearest neighbor search method (as given in Section III) and use them as few shot examples along with the task description.
3) **C-RAIL:** Our proposed contrastive retrieval-augmented in-context learning.

TABLE II
PERFORMANCE ON POSITIVE CLASS ACROSS DATASETS (PRECISION, RECALL, F1).

Dataset	Method	Precision	Recall	F1
Immunotherapy	Zero-shot	17.4	66.7	27.6
	RAIL	26.9	38.8	31.1
	C-RAIL	23.5	66.7	**34.8**
Pima Diabetes	Zero-shot	50.9	77.6	61.4
	RAIL	53.7	67.9	60.0
	C-RAIL	52.0	79.5	**62.8**

D. Results

Table II reports the F1-score on the positive class for both datasets. On the Immunotherapy dataset, C-RAIL improves performance over both zero-shot and standard retrieval-augmented ICL, despite the severe imbalance. The RAIL approach improves precision, but recall is reduced. This occurs because the nearest neighbors retrieved for each input contain many more examples from the negative class than from the positive class, which biases the LLM toward predicting the negative class impacting the overall recall measure for the positive class. The C-RAIL approach mitigates this issue by balancing the retrieved examples, resulting in improved precision while maintaining the same recall as the zero-shot approach, thereby leading to overall gains. We see similar trends on the Pima Diabetes dataset. The C-RAIL method achieves the best performance, indicating the benefit of balanced contrastive retrieval even in moderately imbalanced clinical data. These results highlight the robustness of C-RAIL in addressing class imbalance in medical classification tasks.

V. CONCLUSION AND FUTURE WORK

We presented a method to determine a set of few shot examples that are used along with the task instructions and input data to an LLM. The method retrieves examples that are most similar to given input for the positive as well as the negative class and uses them in the prompt. This mitigates the class imbalance problem in the few shot examples. Our experimental results show that the proposed method improves overall performance. In our future work, we plan to experiment with better embedding based nearest neighbor approaches and see how it effects the classification performance. We also plan to explore *thinking* LLM models on top of retrieved contrastive examples near the input data to generate explanations for conditions under which outcomes are changed and see how that effects the overall classification performance.

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